Supplementary Material
Measuring spiral structure and testing its incremental information for galaxy redshift estimation

Nalin Dhiman

Abstract

This supplement provides the measurement definitions, reference conventions, learning design and numerical details needed to interpret and reproduce the main results. It also develops the physical boundary between image-based redshift prediction and dynamical distance inference. The finite-equilibrium and Tully–Fisher calculations are supporting analyses with independent kinematic inputs; they are not features of the photometric-redshift model.

Contents

S1 Samples, identities and analysis units . . . 1
S2 Coordinates, masks and photometric conventions . . . 2
S2.1 Native sampling and annotation projection . . . 2
S2.2 Background and radial structure . . . 2
S3 PSF matching and noise propagation . . . 3
S4 Exact local feature representation . . . 3
S5 Reference classes, learner candidates and calibration . . . 4
S6 Downstream features and cross-fitting . . . 5
S7 Group-conformal interval construction . . . 6
S8 Effect sizes, estimands and prospective planning . . . 6
S8.1 Literature-motivated observable controls . . . 7
S9 Centerline validation and reference uncertainty . . . 7
S10 Archival published-pitch comparison . . . 8
S11 Physical sizes, arm widths and error propagation . . . 8
S12 A globally finite dynamical equilibrium . . . 9
S12.1 Distribution function and pressure moment . . . 9
S12.2 Flattened baryons and isolated gravity . . . 9
S12.3 Numerical validation and three-galaxy application . . . 10
S12.4 Distance scaling and why viscosity is not measured . . . 11
S13 A calibrated kinematic distance route . . . 11
S14 Reproducibility, outputs and interpretation . . . 12

S1 Samples, identities and analysis units

The principal annotation source is Galaxy Zoo:3D [Masters et al., 2021], combined with native Legacy Survey DR9 science images [Dey et al., 2019]. Observed catalog features and the inherited heliocentric `NSA_Z` label come from official MaNGA DRPall within SDSS-IV DR17 [York et al., 2000, Bundy et al., 2015, Abdurro'uf et al., 2022]. Identity joins precede feature assembly, so incidental image-download or row order cannot assign one galaxy's features to another galaxy's target. No physical radius, catalog stellar mass, mass inferred from a rotation curve or reference distance enters the redshift design matrix.

The initial strong-spiral cohort has 160 objects with a spiral-vote fraction of at least 0.7 and the stipulated

single-center selection. It excludes 32 objects used for earlier feasibility work. Tile membership is

$$g(\alpha, \delta) = \left(\left\lfloor \frac{\alpha}{10^\circ} \right\rfloor, \left\lfloor \frac{\delta + 90^\circ}{10^\circ} \right\rfloor \right), \tag{S1}$$

with right ascension on its standard wrap. Each tile belongs to one split. The catalog split has 91/40/29 galaxies and 38/13/13 sky groups for training/calibration/test. The separate additional acquisition excludes all 192 earlier identities and all 67 earlier tiles, leaving 99 objects in 39 new groups. All 59 eligible strong spirals in that acquisition form the principal additional cohort, in 30 groups. The other 40 objects describe weaker or uncertain morphology and are not verified arm-free negatives.

The units of analysis differ by question. For region AP the basic outcome is one score per galaxy. For the primary redshift contrast it is a mean paired effect within each sky group. For centerline validation it is an entire galaxy with its curve sets. None treats correlated pixels as independent galaxies. Hash-based selection and spatial blocking make a pilot reproducible; they do not yield a volume-complete or survey-representative sample. Blocking is a general response to dependence [Roberts et al., 2017], whose residual strength must still be examined in a future larger study.

Table S1: Coverage denominators. "Defined AP" requires valid scored pixels of both reference classes. The additional redshift analysis retains the missing-image object through training-fitted imputation.

Cohort	Objects	Sky groups	Defined arm AP
Original training	91	38	training task
Original calibration	40	13	calibration task
Original test, three votes	29	13	29
Additional strong spirals, three votes	59	30	57
Additional strong spirals, five votes	59	30	53
Separate geometry development/test	12/12	not the same split	centerline task

S2 Coordinates, masks and photometric conventions

S2.1 Native sampling and annotation projection

The science grid is the native common r-band Legacy grid, approximately 0.262 arcsec per pixel. Science and inverse-variance planes must have matching band, brick, dimensions and celestial WCS; all three bands must share the registered grid and contain the requested galaxy center. Coordinate consistency is checked on a 3×3 grid including corners, edges and center. Nonfinite coordinates or an excessive interband displacement cause an explicit failure. An image extension is selected from audited metadata, rather than by assuming the last FITS extension contains the target.

The Galaxy Zoo:3D reference masks retain their original annotation coordinate system. For each native science pixel, WCS transforms its sky coordinate to the annotation grid and nearest-neighbor sampling transfers the vote count. The science flux is not resampled in this primary pipeline. Nearest-neighbor projection is suitable for preserving discrete label semantics, but does not create information at the finer native sampling. Comparisons remain limited by the reference resolution.

The allowed science region is a circular 3–23 arcsec annulus about the supplied human center. Any positive foreground-star reference masks that pixel in every band. Invalid inverse variance, nonfinite pixels and array boundaries are also excluded. One common feature mask is used across every band and scale. Reference coverage additionally limits *scoring* and supervised pixel sampling; it does not limit the image-defined annulus used to form downstream redshift summaries.

S2.2 Background and radial structure

A border of width $\max[2, \lfloor 0.12 \min(N_x, N_y) \rfloor]$ pixels supplies a robust background estimate. Up to five iterations remove values farther than 3.5 robust standard deviations from the median. The robust scale is

$$\widehat{\sigma}_{\mathrm{bg}} = 1.4826 \ \mathrm{median}\, |I - \mathrm{median}(I)|. \tag{S2}$$

This statistic measures local image scatter, not a complete spatial noise covariance. Extended galaxy light can contaminate a small cutout's border.

The radial profile is the median in one-pixel-wide radial bins, using valid pixels and requiring at least four values in a bin. Missing bins are interpolated and the profile is smoothed with a one-bin Gaussian. Subtracting this profile yields the residual J. A nearest-valid fill supplies values needed to evaluate convolutions internally; filled pixels and their guarded neighborhoods never become accepted feature pixels. For the original supervised region experiment the radial coordinate is projected and circular. An elliptical profile is used only when an independently specified disk geometry is supplied for the distinct geometric analysis.

Catalog fluxes in nanomaggies become magnitudes via

$$m_b = 22.5 - 2.5 \log_{10} F_b, \qquad \sigma_{m_b} \simeq \frac{2.5}{\ln 10} \frac{\sigma_{F_b}}{F_b}. \tag{S3}$$

Nonpositive or unavailable catalog fluxes become missing magnitude values, handled by training-only imputation. Pixel color descriptors instead use signed asinh differences, retaining negative noisy fluxes. An annular "positive flux" is explicitly the sum of $\max(I - B, 0)$ over valid pixels; it is an image descriptor, not a claim of unbiased total photometry.

S3 PSF matching and noise propagation

Let the circular moment-equivalent Gaussian width in band b be w_b in FWHM units. Matching to $w_* = \max_b w_b$ uses

$$\sigma_{k,b} = \frac{\sqrt{w_*^2 - w_b^2}}{2\sqrt{2 \ln 2}}, \tag{S4}$$

in angular units. No sharpening is performed. For a normalized kernel K and validity indicator M, masked convolution is

$$\widetilde{I}_j = \frac{\sum_k K_{jk} M_k I_k}{a_j}, \qquad a_j = \sum_k K_{jk} M_k. \tag{S5}$$

Assuming independent input noise, the propagated diagonal variance is

$$\widetilde{V}_j = \frac{\sum_k K_{jk}^2 M_k V_k}{a_j^2}. \tag{S6}$$

This requires the *squared* kernel. It does not make the smoothed pixels independent. Their covariance is

$$\mathrm{Cov}(\widetilde{I}_j, \widetilde{I}_l) = \frac{\sum_k K_{jk} K_{lk} M_k V_k}{a_j a_l} \tag{S7}$$

under the same idealization. Native coadds may already have covariance not represented in the inverse-variance plane.

Pixels with convolution support below 0.99 are excluded. After this matching, the feature mask also requires more than four times the largest feature σ from holes and borders: the 1.6 arcsec filter therefore requires a 6.4 arcsec guard. At least 100 common valid feature pixels are required. Coverage is measured after these guards, rather than silently filling masked areas with model predictions. Bands are not coadded for the primary classifier; each band retains its own feature block.

A deterministic diagnostic using the first development galaxy illustrates the difference between moment matching and complete PSF matching. The three matched moment-equivalent FWHM values agree to approximately 10^{-5} in relative spread, but their positive-stamp half-light radii still span 0.636–0.735 arcsec and their 80%-light radii 1.139–1.299 arcsec. Approximate matching therefore controls one resolution statistic without equalizing the core or wings. This one-object diagnostic is a method check, not a population error bound or a basis for retuning the frozen test analysis.

S4 Exact local feature representation

Write $N = I - B$ for the background-subtracted native signal, $J = I - m(R)$ for the radial residual, and $S(X) = \max[1.4826\,\mathrm{MAD}(X), 0.1\,\mathrm{sd}(X), 10^{-12}]$ for the stabilizing scale on common feature pixels. Let $s_N = S(N)$, $s_J = S(J)$ and $f = \max(\widehat{\sigma}_{\mathrm{bg}}, 0.1 s_N, 10^{-12})$. The brightness median is μ_N. The filter standard deviations are $\sigma_1 = 0.4$, $\sigma_2 = 0.8$, $\sigma_3 = 1.6$ arcsec, converted to pixels before discrete convolution. Derivative factors use pixel-coordinate σ consistently.

For each band let $L_k = G_{\sigma_k} * J$ and

$$H = \sigma_2^2 \begin{pmatrix} \partial_{xx} L_2 & \partial_{xy} L_2 \\ \partial_{xy} L_2 & \partial_{yy} L_2 \end{pmatrix}. \tag{S8}$$

Order its eigenvalues by absolute magnitude, $|\lambda_s| \le |\lambda_l|$. The bright-ridge strength is

$$\mathcal{R} = \max(-\lambda_l, 0) \exp\left[-\frac{1}{2} \left(\frac{\lambda_s}{0.5 \max(|\lambda_l|, 10^{-12})} \right)^2 \right]. \tag{S9}$$

It is a local descriptor inspired by established Hessian approaches [Frangi et al., 1998], not a new physical definition of an arm. Scale-normalized filtering is related to the broader scale-space framework of Lindeberg [1998]; this experiment uses three fixed scales rather than automatic physical-scale selection.

Table S2: Thirteen features repeated in each of g, r, z. Together with three radius terms and two color terms they give 44 features. All are computed without redshift or physical-distance information.

Feature	Definition or meaning
Signed intensity	$\operatorname{asinh}(N/f)$
Normalized brightness	$(N - \mu_N)/s_N$
Radial contrast	J/s_J
Radial profile	$(N - J)/s_N$
Fine/middle contrast difference	$(L_1 - L_2)/s_J$
Middle/coarse contrast difference	$(L_2 - L_3)/s_J$
Negative Laplacian-like response	$-\operatorname{tr} H/s_J$
Bright ridge	$\mathcal{R}/s_J$
Gradient strength	$\sigma_1 \lvert \nabla L_1 \rvert / s_J$
Fine residual	L_1/s_J
Coarse residual	L_3/s_J
Radial orientation, even	$(\mathcal{R}/s_J) \cos 2(\alpha - \phi_{\mathrm{sky}})$
Radial orientation, odd	$(\mathcal{R}/s_J) \sin 2(\alpha - \phi_{\mathrm{sky}})$

Here α denotes the local bright-ridge tangent orientation and ϕ_{sky} the projected center-to-pixel angle. Double angles remove the arbitrary sign of an eigenvector; these features do not measure a globally deprojected spiral pitch. The three radius terms are u, u^2 and $\log(1+u)$, with $u = R_\theta/23$ arcsec. The two color terms are

$$C_{ab} = \operatorname{asinh}(N_a/f_{ab}) - \operatorname{asinh}(N_b/f_{ab}), \qquad f_{ab} = \max(\widehat{\sigma}_a, \widehat{\sigma}_b, 10^{-12}), \tag{S10}$$

for $(a, b) = (g, r), (r, z)$ in common declared photometric units. This signed color is useful for noisy pixels but is not a standard magnitude difference.

S5 Reference classes, learner candidates and calibration

The primary binary label is $y_j = \mathbf{1}[v_j \ge 3]$, where v_j is the number of human arm polygons covering the scored pixel. The classifier is trained against this operational definition. Because the reference is incomplete, $y = 0$ denotes the comparison class under the task, not verified physical absence. The selection already favors strong spirals; no real-negative specificity is inferred from low-vote or unasked objects.

Each training galaxy supplies the same number of randomly selected positive and comparison pixels, $\min(400, n_+, n_-)$. Sampling uses a deterministic identity-derived seed and inverse sample-size weights give each usable galaxy the same total weight. Candidate evaluation averages per-galaxy AP within three-fold training sky-group cross-validation. The seven candidates are: brightness/radius logistic regression; brightness/radius ExtraTrees at depth 8 or 12; full logistic regression at $C = 0.1$ or 1; and full ExtraTrees at depth 8 or 12. Standardization for logistic regression is fitted within the training fold. Tree models use 80 trees, leaf size 32, feature fraction 0.7 and a fixed seed. The randomized tree construction follows Geurts et al. [2006] through scikit-learn [Pedregosa et al., 2011].

The selected full and brightness/radius models both have depth 12. Their training-CV AP values are 0.6757 and 0.6335. Thus the original test comparison is between independently training-selected models with a common eligible capacity family, not between a complex learner and an intentionally weak baseline. Pixel-class balancing does change score prevalence and motivates distinguishing raw ranking scores from calibrated annotation probabilities.

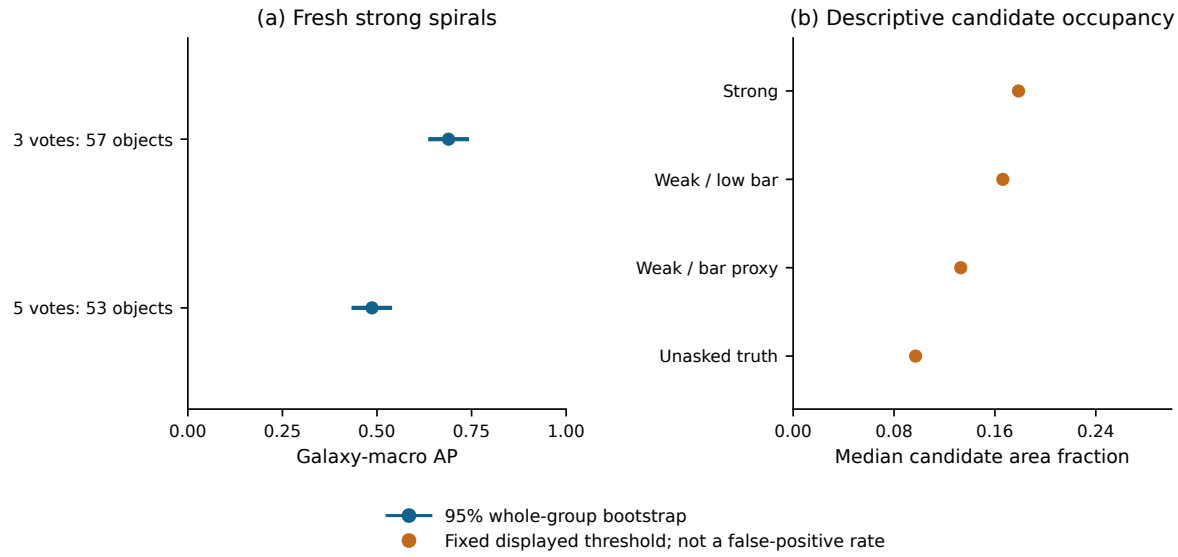


Figure S1: Sensitivity to the human consensus threshold. Three- and five-vote targets have different positive prevalence and evaluable samples, so changes in AP cannot be interpreted as a controlled loss of physical arm accuracy. Low-vote/unasked regions are not promoted to verified negatives.

The display mapping takes raw score s to

$$p_{\text{display}} = \text{logistic}[a\,\text{logit}(\text{clip}(s)) + b], \qquad a > 0. \tag{S11}$$

This is a sigmoid calibration [Platt, 2000]; the fitted values are $a = 0.81289$, $b = -1.48564$. The calibration-selected display threshold is 0.30. The positive slope preserves AP ranking. A different artifact retains raw s and fixed threshold 0.5 for downstream inference, so these calibration labels cannot influence the redshift representation.

For ranked pixels with recall increments $R_k - R_{k-1}$ and precision P_k,

$$\text{AP} = \sum_k (R_k - R_{k-1}) P_k. \tag{S12}$$

AP depends on reference prevalence as well as ranking. A threshold change from three to five polygons changes prevalence and occasionally eliminates a scoring class; it is not a fixed-task perturbation alone. On the original test, five-vote AP is 0.5012 on 28 objects; on the additional strong cohort it is 0.4868 on 53 objects. The corresponding primary three-vote AP values are 0.7197 on 29 and 0.6890 on 57. These denominators are part of the result.

S6 Downstream features and cross-fitting

Training arm summaries are out-of-fold by sky group. If $h^{(-k)}$ is the classifier fitted without fold k and $\mathcal{A}$ is the image-defined annulus, a training galaxy in that fold receives

$$A_i = \mathcal{T}\{h^{(-k)}(X_{ij}) : j \in \mathcal{A}_i\}. \tag{S13}$$

Calibration and test galaxies instead receive $\mathcal{T}$ applied to the single full-training classifier. This is cross-fitting of the arm representation. Upstream candidate choice uses the training set, so downstream training-CV results conditional on these cached descriptors are not a fully nested assessment of the complete model-selection process. The original final test groups remain outside all training selection.

For raw scores s_j and normalized radii u_j, define

$$\bar{s} = n^{-1} \sum_j s_j, \qquad u_s = \frac{\sum_j s_j u_j}{\sum_j s_j}, \tag{S14}$$

$$H_s = -n^{-1} \sum_j [s_j \log_2 s_j + (1 - s_j) \log_2 (1 - s_j)], \qquad f_s = n^{-1} \sum_j \mathbf{1}[s_j \geq 0.5]. \tag{S15}$$

Together with $Q_{0.9}(s)$, sd(s) and inner/outer score means at $u = 0.55$, these give eight summaries. Zero-weight and empty subregion conventions are fixed by the implementation. Entropy here is a feature of the score distribution; it is not a posterior uncertainty on physical arms.

The four regressor families are nested in their principal information blocks: P, PG, PGI and PGIA. P has five magnitudes and adjacent colors. PG adds angular half-light radius and axis ratio. PGI adds three annular positive-flux sums, the r-band half-positive-flux radius, r residual contrast, and two mean signed colors. PGIA adds A_i. All share five magnitude errors, three PSF widths, three native noise estimates and an arm-feature availability flag. Training-median imputation, including missingness indicators, is fitted within training. No test value determines an imputation median.

The target transformation $t = \log(1 + z)$ preserves order and compresses the redshift range. It does not cause the regressor to learn a cosmological distance. Predictions $\hat{z} = \exp(\hat{t}) - 1$ are assessed against inherited spectroscopic catalog labels. Redundant magnitudes and colors are intentionally kept fixed across ablations; the goal is a controlled contrast, not a claim that this small ExtraTrees system is an optimal photometric-redshift architecture. Image-based CNN inference and larger morphology-assisted methods provide relevant alternatives [Pasquet et al., 2019, Soo et al., 2018].

S7 Group-conformal interval construction

Let $\widehat{t_i}$ be the prediction from a model fitted only to training galaxies. Each calibration group contributes

$$S_g = \max_{i \in g} |t_i - \widehat{t_i}|. \tag{S16}$$

For G_c calibration groups and miscoverage α, define

$$k = \lceil (G_c + 1)(1 - \alpha) \rceil, \qquad q = S_{(k)}, \tag{S17}$$

where $S_{(k)}$ is the kth ordered group score; a rank beyond G_c requires an infinite interval. Here $G_c = 13$, $\alpha = 0.1$ and $k = 13$. Prediction intervals are transformed monotonically:

$$[z_-, z_+] = [\max(0, \exp(\widehat{t} - q) - 1), \exp(\widehat{t} + q) - 1]. \tag{S18}$$

Under exchangeability of the calibration and future group objects, the rank argument supplies coverage for the group maximum, hence simultaneous coverage of all retained members of a new comparable group. It does not give precise conditional coverage at each feature value. The basic split-conformal theory is described by Lei et al. [2018]; Angelopoulos and Bates [2021] gives a practical account. Our calculation does not implement the likelihood-ratio correction that can address a specified covariate shift [Tibshirani et al., 2019].

The calibrated set is small, so the maximum residual can produce broad intervals. Original P coverage 29/29 and additional P coverage 57/59, with median widths 0.08954 and 0.09043, are empirical diagnostics, not evidence of precise physical distances. The additional cohort has whole-group coverage 28/30. Its selection and missingness can differ from the calibration groups; coverage is therefore measured rather than assumed to transfer.

S8 Effect sizes, estimands and prospective planning

The primary paired effect averages absolute-error differences within each sky group and then gives groups equal weight. If group sizes differ,

$$\frac{1}{G} \sum_g \frac{1}{n_g} \sum_{i \in g} d_i \neq \frac{1}{N} \sum_i d_i \tag{S19}$$

in general. The same distinction applies to paired AP improvements. We report galaxy-macro measurements for descriptive performance and equal-group paired effects for the central comparison. Percentile intervals use whole-group bootstrap resampling with 100,000 draws for the final redshift contrasts [Efron, 1979]. They condition on the fitted models and cohorts.

The practical margin $\delta_z = 0.001$ was introduced retrospectively. A second sensitivity uses $\delta_t = 0.00138663$ in $t = \log(1 + z)$, equal to 10% of a specified training-only PGI CV error on that transformed scale. It must not be mislabeled as a redshift margin. Equivalence testing reverses the ordinary null: it seeks to exclude effects outside a prespecified interval, rather than merely failing to reject zero [Schuirmann, 1987]. No primary useful-gain or equivalence conclusion is established here. The secondary transformed-scale margin is more permissive and remains a sensitivity, not a newly confirmatory success.

Table S3: Principal equal-group redshift contrast. Positive values favor arms. The Student-t interval is a sensitivity to small group counts, not an independent replication.

Cohort	Effect	Bootstrap 95%	Student-t 95%
Original, $G = 13$	-0.000471	$[-0.001391, 0.000367]$	$[-0.001491, 0.000549]$
Additional, $G = 30$	0.000440	$[-0.000555, 0.001452]$	$[-0.000620, 0.001500]$

For a frozen future comparison with independent approximately normal group effects of standard deviation s_d, the usual planning scale for a two-sided test against zero is

$$G \approx \frac{(z_{1-\alpha/2} + z_{1-\beta})^2 s_d^2}{\delta_z^2}. \tag{S20}$$

The numerical planning uses a noncentral-t calculation instead of this large-sample approximation. With $s_d = 0.00283859$, the larger observed cohort SD, detecting a true gain 0.001 requires 66 groups at 80% power and 87 at 90%. With the normal-assumption upper-variance scenario $s_d = 0.00363255$, these become 106 and 141 groups. A test of superiority over 0.001 when the true gain is 0.002, or an equivalence design around zero, is a different calculation. These figures are planning assumptions, not estimates of observed power or predictions that morphology will help.

S8.1 Literature-motivated observable controls

The L5 control supplements P and apparent geometry with observed r-band surface brightness and catalog Sérsic index. Surface brightness is

$$\mu_r = m_r + 2.5 \log_{10}(2\pi q R_{50}^2), \tag{S21}$$

with R_{50} in arcsec and apparent axis ratio q. This observed quantity uses no redshift-based dimming or rest-frame correction. L3 restricts explicit photometric and magnitude-error predictors to grz, with the stated apparent geometry and grz quality information retained. Appending A gives L5A/L3A. These are common-capacity controls motivated by Soo et al. [2018]; they do not reproduce that paper's neural architecture or CS82 sample.

Table S4: Retrospective controls on the same 59 additional strong spirals. All are fitted only on the original 91 training galaxies. Group and galaxy estimands can change in different directions, as for L3/L3A.

Family	Galaxy MAE	Equal-group MAE
L5	0.012463	0.013132
L5A	0.012393	0.012547
L3	0.012504	0.012928
L3A	0.012634	0.012786

The equal-group L5 gain is 0.000584 with 95% interval $[-0.000331, 0.001658]$; the L3 gain is 0.000142 with interval $[-0.000802, 0.001195]$. No test-based deployed-model reselection or new interval-calibration claim is made for these additional baselines.

S9 Centerline validation and reference uncertainty

The geometry sample contains 24 Galaxy Builder subjects with human freehand traces, author-cleaned clustered arms and fitted disk geometry [Lingard et al., 2020, 2021]. The SDSS r image arrays have normalized linear values, PSFs and 0.396 arcsec per pixel metadata. They are not absolutely calibrated three-band flux images. Source image centers exclude earlier objects within 10 arcsec, but the geometry sample is not a new survey or a tile-disjoint extension of the principal experiment.

Twelve development objects choose one smoothing/template configuration. The extractor searches logarithmic-spiral contrast templates after deprojection with the supplied geometry. Twelve test objects evaluate the fixed configuration. Source references were available during preparation; there was no sealed external test. Scores were not used to tune the configuration after the test evaluation. This distinction matters when planning a future confirmatory geometry benchmark.

Let P and T be sampled predicted and reference centerline point sets on common annular support, and let $d(p,T) = \min_{t\in T} |p-t|$. At angular tolerance ϵ the point-based metrics are

$$\mathrm{Prec}_\epsilon = \frac{1}{|P|}\sum_{p\in P} \mathbf{1}[d(p,T)\le\epsilon], \quad \mathrm{Rec}_\epsilon = \frac{1}{|T|}\sum_{t\in T} \mathbf{1}[d(t,P)\le\epsilon], \tag{S22}$$

with $F1 = 2\,\mathrm{Prec}\,\mathrm{Rec}\,/(\mathrm{Prec}+\mathrm{Rec})$. Curves are sampled approximately uniformly in projected angular length at 0.1 arcsec spacing, with at least eight points per curve. The main tolerance is the supplied PSF FWHM; half that tolerance is a sensitivity. One-to-one segment assignment minimizes the symmetric median point-to-curve distance. Only pairs with assigned distance at most twice the PSF FWHM enter pitch comparison, preserving the matched denominator. Test F1 is 0.223 (95% whole-galaxy interval 0.105–0.353); at half-PSF it is 0.135 (0.049–0.235). There are 70 predicted and 19 reference segments, with seven matched segments whose median absolute pitch difference is 11.42 degrees. A conditional pitch statistic cannot describe every unmatched arm or galaxy.

Twenty of 24 synthetic nonspiral images produce at least one false curve. This is a controlled stress test of the template behavior, not a measured false-positive rate for a real observed nonspiral population. Human traces also have uncertainty: different volunteers can select different arm pieces, and author aggregation/fitting imposes a specific reference convention. Independent tracing is stronger than visual plausibility, but not error-free physical truth. No observed arm-width accuracy is established.

S10 Archival published-pitch comparison

The Hart et al. [2017] author catalog contains 6,222 parent galaxies, with SpArcFiRe-based pitches after reliable-arc selection. Its exact DR7 identifiers match all 24 Galaxy Builder subjects, with maximum coordinate separation 0.020 arcsec. Thirteen pitches are finite and eleven retain the published -999 sentinel as missing. There are seven measured development and six measured test objects. No missing pitch is scored as zero or as absence of arms.

The Galaxy Builder reference summary is an unsigned pitch averaged over author-accepted arm fits using their stored *projected native-image* lengths. The archival SpArcFiRe summary uses selected reliable arcs with a different standardized/deprojected weighting convention and chirality selection. Equal galaxy summaries need not imply agreement about individual arm locations or completeness. The arithmetic therefore measures a convention-dependent catalog difference.

For the six measured test identities the mean absolute difference is 7.727 degrees, median absolute difference 7.280 degrees, and the conditional whole-galaxy bootstrap interval for mean absolute difference is 5.061–10.685 degrees. On those same identities, a fixed development-median constant has mean absolute difference 7.122 degrees. Seven development identities give mean absolute difference 10.357 degrees; their signed mean is 6.925 degrees. These different statistics are not interchangeable.

This is not an execution of the author runtime on the present images. The identities share the Hart parent population and individual classifier training membership is not established. The comparison is retrospective and does not create an independent population test. It neither rescues the failed centerline prototype nor establishes superiority over SpArcFiRe [Davis and Hayes, 2014, Hart et al., 2017]. A future algorithm comparison must harmonize input images, support, arc-selection definitions, failure handling and training-overlap exclusions.

S11 Physical sizes, arm widths and error propagation

For supplied external D_A, angular separations convert through $L = D_A\theta$. First-order uncertainty, including a possible covariance, is

$$\mathrm{Var}(L) \simeq \theta^2\,\mathrm{Var}(D_A) + D_A^2\,\mathrm{Var}(\theta) + 2D_A\theta\,\mathrm{Cov}(D_A,\theta). \tag{S23}$$

A physical size reported without the supplied distance uncertainty can appear more informative than it is. Disk deprojection adds uncertainty in center, inclination and position angle. For example, $Y = y'/\cos i$ gives $\partial Y/\partial i = y'\sec i\tan i$, exposing rapidly increasing sensitivity as the disk approaches edge-on. A projected separation between two galaxies is also not a three-dimensional spatial separation without line-of-sight information.

For approximately Gaussian transverse profiles one sometimes writes $w_{\mathrm{obs}}^2 \simeq w_{\mathrm{int}}^2 + w_{\mathrm{PSF}}^2$ in a common width convention. Deconvolution near $w_{\mathrm{obs}} = w_{\mathrm{PSF}}$ is unstable and requires uncertainty propagation, background control, a defensible local cross-arm direction and a PSF model. Real arm profiles and PSFs need not be

Gaussian. Conditional fits around supplied reference curves can be useful diagnostics, but they are not independent observed-width validation and are not accepted distance predictors in this study.

S12 A globally finite dynamical equilibrium

This supporting calculation asks whether the dynamical component can be formulated consistently. It uses known equilibrium physics, not an additional image-derived distance calibration. Pressure support in a stationary, nonrotating isotropic halo satisfies

$$\nabla P = -\rho \nabla \Phi, \qquad \nabla^2 \Phi_h = 4\pi G \rho, \qquad \Phi = \Phi_h + \Phi_b. \tag{S24}$$

An untruncated isothermal relation $P = \rho s^2$ gives $\rho \propto \exp(-\Phi/s^2)$ but does not by itself specify a globally finite isolated halo. Cutting such a profile at an arbitrary nonzero-pressure edge requires a boundary stress. We instead implement a lowered-isothermal closure with a finite zero-pressure surface, following the established King family [King, 1966, Gieles and Zocchi, 2015]. Self-consistent disk/halo construction also has a long history [Kuijken and Dubinski, 1995].

S12.1 Distribution function and pressure moment

Let $E = v^2/2 + \Phi$, choose a cutoff energy E_t, and define

$$f(E) = \begin{cases} A\{\exp[(E_t - E)/s^2] - 1\}, & E < E_t, \\ 0, & E \geq E_t. \end{cases} \tag{S25}$$

An ergodic function of conserved energy is a stationary collisionless distribution in the prescribed static potential. Its velocity moments also give an isotropic Jeans/barotropic equilibrium. Neither interpretation requires a collisional viscosity. Set $W = (E_t - \Phi)/s^2$ and define, for $W > 0$,

$$D(W) = e^W \operatorname{erf}\sqrt{W} - 2\sqrt{W/\pi}\left(1 + \frac{2W}{3}\right), \tag{S26}$$

$$Q(W) = e^W \operatorname{erf}\sqrt{W} - 2\sqrt{W/\pi}\left(1 + \frac{2W}{3} + \frac{4W^2}{15}\right). \tag{S27}$$

Both are zero for $W \leq 0$. With central density ρ_0 and central binding W_0, the moments are

$$\rho(W) = \rho_0 \frac{D(W)}{D(W_0)}, \qquad P(W) = \rho_0 s^2 \frac{Q(W)}{D(W_0)}. \tag{S28}$$

Since $Q'(W) = D(W)$ and $\nabla W = -\nabla\Phi/s^2$, equation (S24) follows directly. The local one-dimensional dispersion is P/ρ, not the fixed parameter s^2. Near the edge,

$$D(W) \sim \frac{8W^{5/2}}{15\sqrt{\pi}}, \qquad Q(W) \sim \frac{16W^{7/2}}{105\sqrt{\pi}}, \qquad \frac{P}{\rho} \sim \frac{2}{7}s^2 W. \tag{S29}$$

Density and pressure therefore vanish continuously on the cutoff equipotential. The cutoff is a specified model family, not an inference of a cosmological halo edge or formation history.

S12.2 Flattened baryons and isolated gravity

The prescribed baryonic potential is a nonnegative superposition of Miyamoto–Nagai components [Miyamoto and Nagai, 1975],

$$\Phi_b(R, z) = -\sum_j \frac{GM_j}{\sqrt{R^2 + [a_j + \sqrt{z^2 + b_j^2}]^2}}, \qquad M_j \geq 0. \tag{S30}$$

Its equatorial circular-speed contribution is

$$v_{b,j}^2(R) = \frac{GM_j R^2}{[R^2 + (a_j + b_j)^2]^{3/2}}. \tag{S31}$$

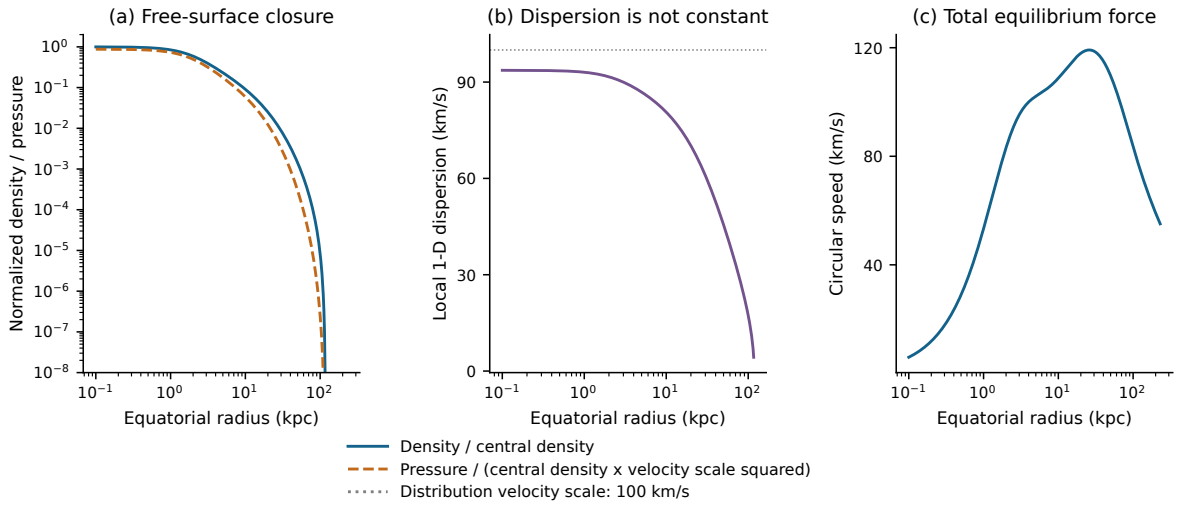


Figure S2: Finite lowered-isothermal equilibrium and numerical closure. The cutoff surface has zero density and pressure; the computational boundary lies in vacuum. This controlled calculation concerns the specified equilibrium family and baryonic potential, not identification of dark-matter microphysics.

Thus equatorial forces determine $a_j + b_j$ more directly than the vertical thickness b_j. The halo density is iterated in the combined potential, so adding a disk force after solving an isolated spherical halo is not mislabeled as exact coupled equilibrium.

For axisymmetric density expanded as $\rho(r,\mu) = \sum_l \rho_l(r) P_l(\mu)$, the isolated Green solution is

$$\Phi_{h,l}(r) = -\frac{4\pi G}{2l+1}\left[r^{-l-1}\int_0^r \rho_l(a)a^{l+2}\,da + r^l\int_r^\infty \rho_l(a)a^{1-l}\,da\right]. \tag{S32}$$

Reflection symmetry retains even l. Quadrature and under-relaxed density iteration close Poisson's equation with the moments above. The numerical sphere must enclose the entire zero-density surface; a state with positive boundary binding is rejected or the domain enlarged. Outside the halo the potential continues through the vacuum multipoles, not an imposed zero-potential wall. The analytic baryon components have finite mass but need not be compactly supported inside the same sphere.

S12.3 Numerical validation and three-galaxy application

An independent spherical ordinary-differential-equation solution checks the zero-baryon limit. The controlled case has maximum relative circular-speed difference 3.55×10^{-5} and total-mass difference 5.36×10^{-5}. The direct pressure-gradient identity has maximum relative residual 1.77×10^{-9}. Increasing the controlled grid from 200 radial points/$l_{\max} = 8$ to 960/$l_{\max} = 32$ reduces the measured Poisson residual from 7.44×10^{-3} to 2.92×10^{-4}; the coarse-to-finest speed difference is 0.0414 km s^{-1}. Doubling the vacuum domain from 250 to 500 kpc changes speed by at most 0.000174 km s^{-1} and mass by 5.43×10^{-6} in relative units. These are discretization checks for controlled inputs, not uncertainty estimates for real galaxies.

For NGC2403, NGC3198 and NGC5055, SPARC supplies external distances, photometric/gas force columns and observed rotation curves [Lelli et al., 2016]. Nonnegative analytic components approximate the baryonic force columns without using observed velocities to define the baryon template. This approximation has baryonic speed errors of several km s^{-1} and cannot reproduce every outward-force feature of a ring-like gas distribution. Published signed gas forces are preserved in the original-force comparison.

A photometric population relation sets

$$h_z = 10^{-0.708}(R_d/\mathrm{kpc})^{0.633}\,\mathrm{kpc}, \tag{S33}$$

related to the disk-flattening information discussed by Bershady et al. [2010] and adopted in the SPARC structural context [Lelli et al., 2016]. It gives 0.241, 0.404 and 0.409 kpc. The stellar analytic model's first absolute vertical moment is matched to this value at $2.2R_d$. This is a population prior, not a direct thickness measurement for each target. A sensitivity varies the adopted height by $e^{\pm 0.25}$ at fixed fitted dark parameters; the largest resulting speed change is 0.494 km s^{-1}. The gas thickness remains an assumed fraction, not a new observation.

The same predeclared image annuli define a regional conditional prediction test: fit velocities outside the annulus and predict inside, using externally supplied distance to convert angles to kpc. These annuli have no independent arm-centerline acceptance and must not be called verified arms. They were already examined in earlier analysis, so the comparison is exploratory. The primary finite coupled model fixes $W_0 = 5$ and fits two dark parameters, ρ_0, s. Table S5 keeps the same three targets and all comparison outcomes. No derivative of observed velocity is used as a predictor of that same velocity.

Table S5: Regional held-out rotation-speed RMSE in $\mathrm{km\,s^{-1}}$. All comparisons condition on external distance and baryonic assumptions. Different baryonic force approximations are explicitly distinguished; the three targets do not establish population superiority.

Halo/force construction	NGC2403	NGC3198	NGC5055
Finite coupled, $W_0 = 5$, analytic baryons	4.69	11.77	18.13
Confined coupled isothermal, same thickness	7.47	3.57	27.90
Finite spherical + analytic baryon forces	8.32	18.01	9.05
Uncontracted isothermal + analytic forces	5.61	5.58	10.44
Uncontracted isothermal + signed SPARC forces	5.64	4.59	6.52

Physical consistency does not guarantee a better conditional prediction. The cutoff also matters: changing W_0 to three or seven at fixed dark parameters can change speeds by tens of $\mathrm{km\,s^{-1}}$. A later training-only cutoff search over three, five, seven and nine reaches endpoints for these targets. This does not identify a unique physical cutoff or establish an observational microphysical inference.

S12.4 Distance scaling and why viscosity is not measured

For free physical mass and length parameters, Newtonian equilibrium admits the similarity

$$r \mapsto \lambda r, \quad M \mapsto \lambda M, \quad \rho \mapsto \lambda^{-2}\rho, \quad \Phi \mapsto \Phi, \quad v_c \mapsto v_c. \tag{S34}$$

The same angular velocities can then correspond to differently scaled systems. The implemented similarity check preserves speed to 1.08×10^{-14} in relative units. This statement concerns freely rescaled mass models. If apparent flux is fixed and a calibrated mass-to-light ratio is supplied, baryonic mass instead scales as D^2; that extra physical calibration and independent kinematics can constrain distance. The similarity must not be used to deny this conditional route.

For a rotating Newtonian fluid the cylindrical shear relevant to viscous angular-momentum transport is

$$\tau_{R\phi} = \eta R \frac{d\Omega}{dR} = \eta \left(\frac{dv_\phi}{dR} - \frac{v_\phi}{R} \right). \tag{S35}$$

Rigid rotation has zero shear; a flat rotation curve has nonzero shear even though $dv_\phi/dR = 0$. An isotropic pressure-supported halo and a gravitational rotation curve do not identify η. A transport claim would require a specified moving fluid, momentum/energy closure, time dependence or radial flow and boundary torque information. The present finite model supplies equilibrium density and gravity, not those missing observations.

S13 A calibrated kinematic distance route

The Tully–Fisher relation adds independent kinematic information to photometry [Tully and Fisher, 1977]. For a baryonic version write

$$y = \log_{10}(M_b/\mathrm{M}_\odot) = ax + b + \epsilon, \qquad x = \log_{10}(V/V_0), \quad \epsilon \sim \mathcal{N}(0, \sigma_{\mathrm{int}}^2). \tag{S36}$$

Characteristic velocity must retain its definition. Flat outer H I speed, optical rotation speed and integrated line width are different observables, as quantified by Lelli et al. [2019]. Calibrated baryonic distances already have an established literature [Schombert et al., 2020]; this supporting implementation is not a new distance law.

Let $C_b = M_b/D^2$ be the distance-free mass/flux coefficient for a fixed stellar mass-to-light convention and gas correction. Then

$$\log_{10}(D/\mathrm{Mpc}) = \frac{1}{2}[ax + b - \log_{10} C_b], \tag{S37}$$

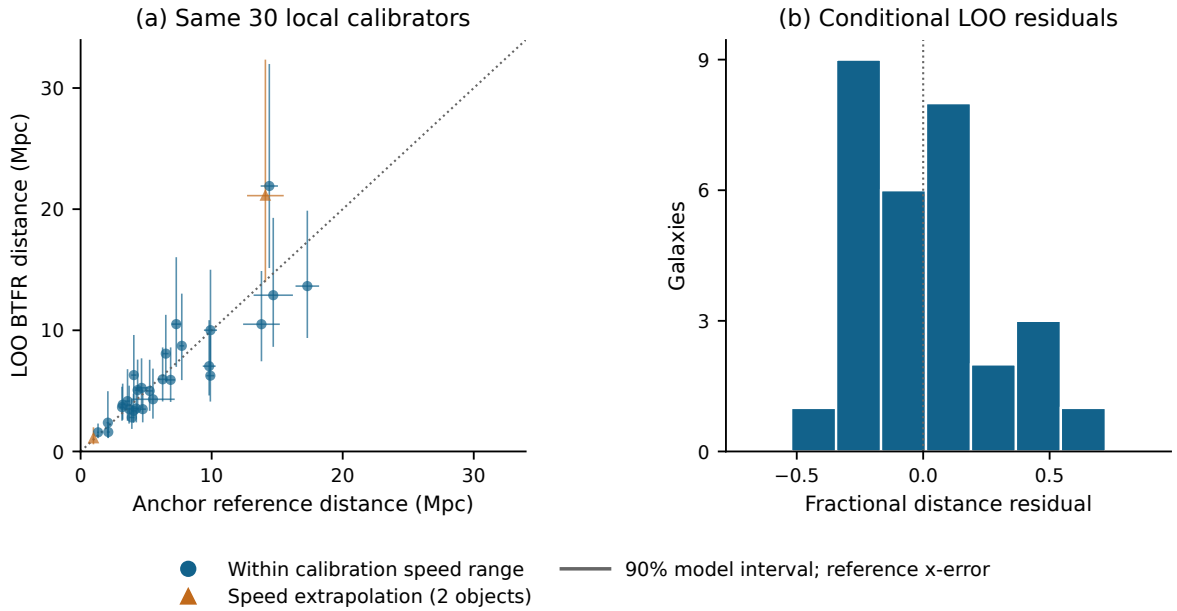


Figure S3: Supporting baryonic Tully–Fisher distance calculation. Distances require matched flux conventions, independent measured kinematics and calibrated anchors. The plotted intervals are model-dependent; the small local leave-one-out sample is not a new blind distance validation.

with unit conventions retained in C_b. Recovering C_b from a published flux-derived mass and its *same reference distance* cancels that distance algebraically; recovering it from inconsistent catalogs would not. Across 153 matched published rows, reconstructed $M_b = 0.5L_{3.6} + 1.33M_{\rm HI}$ agrees with the rounded published masses to at most 0.005 dex. This checks source consistency, not the correctness of their distance anchors. A coherent test $D \mapsto 2.5D$, $M_b \mapsto 6.25M_b$ preserves C_b to floating-point precision.

For Gaussian errors and a flat latent-log-speed approximation, integrating over uncertain x gives a vertical likelihood with variance

$$s_i^2 = \sigma_{y,i}^2 + a^2\sigma_{x,i}^2 + \sigma_{\rm int}^2. \tag{S38}$$

The implementation uses bounded slope and scatter priors and propagates posterior and target flux/speed uncertainty. It does not fit a population selection function or shared anchor covariance. Distance-related mass error and reference-distance error must not be independently counted twice when constructing the flux coefficient.

Thirty calibrators with external stellar-distance reference methods are evaluated leave-one-out. The error-aware model has median absolute fractional distance error 17.87%, compared with 17.60% for ordinary least squares on the same updated data. Its nominal 90% model intervals cover 27/30 reference points, with median width 82.21% of the predicted distance. This is a useful calibrated route, but the broad intervals and small reused anchor sample preclude a claim of new precision or externally calibrated coverage.

A retrospective matched-width audit keeps the same 23 target galaxies and 22 leave-one-out calibrators for all alternatives. Proper line-width calibration gives median absolute error 23.13%, mean absolute error 24.60% and interval coverage 22/23. Deliberately substituting that width into a flat-speed calibration happens to lower median error to 14.82%, but raises mean error to 31.47% and lowers coverage to 18/23. This counterexample shows why choosing a model by one favorable small-sample statistic would be misleading. No winner is selected from the audit.

Six MaNGA Hα velocity maps provide a further data-availability check. All are processed, but none passes the specified quality/extent/flatness requirements for the flat-H I-speed calibration. Distance is therefore withheld. This outcome concerns those optical maps and those gates, not a claim that Hα is universally unsuitable for kinematic distances. Neither these maps nor the three illustrative equilibrium targets supply missing image-only distance labels for the redshift experiment.

S14 Reproducibility, outputs and interpretation

The companion repository provides publicly available source, model metadata, machine-readable measurements and figure provenance: https://github.com/nalin-dhiman/Galaxy-arm-project. Original archive

products are acquired using their credited public interfaces. The reproducibility sequence is to acquire/verify identities and native data; validate coordinates, PSFs and coverage; apply the frozen region model; assemble observable features by identity; apply the fixed redshift regressors and calibration; and compute galaxy/group outcomes with the declared denominators. A fresh scientific experiment requires a new fixed sample and protocol rather than overwriting the reported frozen evaluation.

Per-object region outputs retain the valid mask, score map, thresholded display map, angular sampling, coverage, feature schema and model identity. Redshift outputs retain the predicted catalog redshift, interval endpoints, feature-family identity, missing-input flags and support status. Accepted science results and diagnostic candidates have separate meanings. An out-of-range or incompatible input is reported explicitly rather than silently converted into a physical distance.

Geometry outputs retain projected/deprojected units, supplied center and disk geometry, candidate versus matched segment counts and the reference convention. Physical conversions retain the supplied distance and its origin. Dynamical outputs retain baryonic assumptions, boundary/Poisson diagnostics and the kinematic observable definition. These records are necessary to prevent a good image visualization, a successful numerical equilibrium or a wide covered redshift interval from being mistaken for a validated absolute-distance measurement.

References

Abdurro'uf, Kathleen Accetta, Conny Aerts, et al. The seventeenth data release of the sloan digital sky surveys: Complete release of MaNGA, MaStar, and APOGEE-2 data. *The Astrophysical Journal Supplement Series*, 259:35, 2022. doi: 10.3847/1538-4365/ac4414. URL https://arxiv.org/abs/2112.02026.

Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification, 2021. URL https://arxiv.org/abs/2107.07511.

Matthew A. Bershady, Marc A. W. Verheijen, Kyle B. Westfall, David R. Andersen, Rob A. Swaters, and Thomas Martinsson. The DiskMass survey. II. error budget. *The Astrophysical Journal*, 716:234–268, 2010. doi: 10.1088/0004-637X/716/1/234. URL https://arxiv.org/abs/1004.5043.

Kevin Bundy, Matthew A. Bershady, David R. Law, et al. Overview of the SDSS-IV MaNGA survey: Mapping nearby galaxies at apache point observatory. *The Astrophysical Journal*, 798:7, 2015. doi: 10.1088/0004-637X/798/1/7. URL https://arxiv.org/abs/1412.1482.

Darren R. Davis and Wayne B. Hayes. Scalable automated detection of spiral galaxy arm segments. *The Astrophysical Journal*, 790:87, 2014. doi: 10.1088/0004-637X/790/2/87. URL https://arxiv.org/abs/1402.1910.

Arjun Dey, David J. Schlegel, Dustin Lang, et al. Overview of the DESI legacy imaging surveys. *The Astronomical Journal*, 157:168, 2019. doi: 10.3847/1538-3881/ab089d. URL https://arxiv.org/abs/1804.08657.

Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7:1–26, 1979. doi: 10.1214/aos/1176344552. URL https://doi.org/10.1214/aos/1176344552.

Alejandro F. Frangi, Wiro J. Niessen, Koen L. Vincken, and Max A. Viergever. Multiscale vessel enhancement filtering. In *Medical Image Computing and Computer-Assisted Intervention—MICCAI'98*, volume 1496 of *Lecture Notes in Computer Science*, pages 130–137, 1998. doi: 10.1007/BFb0056195. URL https://link.springer.com/chapter/10.1007/BFb0056195.

Pierre Geurts, Damien Ernst, and Louis Wehenkel. Extremely randomized trees. *Machine Learning*, 63:3–42, 2006. doi: 10.1007/s10994-006-6226-1. URL https://link.springer.com/article/10.1007/s10994-006-6226-1.

Mark Gieles and Alice Zocchi. A family of lowered isothermal models. *Monthly Notices of the Royal Astronomical Society*, 454:576–592, 2015. doi: 10.1093/mnras/stv1848. URL https://arxiv.org/abs/1508.02120.

Ross E. Hart, Steven P. Bamford, Wayne B. Hayes, et al. Galaxy zoo and SpArcFiRe: constraints on spiral arm formation mechanisms from spiral arm number and pitch angles. *Monthly Notices of the Royal Astronomical Society*, 472:2263–2279, 2017. doi: 10.1093/mnras/stx2137. URL https://arxiv.org/abs/1708.04628.

Ivan R. King. The structure of star clusters. III. some simple dynamical models. *The Astronomical Journal*, 71:64–75, 1966. doi: 10.1086/109857. URL https://articles.adsabs.harvard.edu/pdf/1966AJ.....71...64K.

Konrad Kuijken and John Dubinski. Nearly self-consistent disk-bulge-halo models for galaxies. *Monthly Notices of the Royal Astronomical Society*, 277:1341–1353, 1995. doi: 10.1093/mnras/277.4.1341. URL https://arxiv.org/abs/astro-ph/9502051.

Jing Lei, Max G'Sell, Alessandro Rinaldo, Ryan J. Tibshirani, and Larry Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113:1094–1111, 2018. doi: 10.1080/01621459.2017.1307116. URL https://arxiv.org/abs/1604.04173.

Federico Lelli, Stacy S. McGaugh, and James M. Schombert. SPARC: Mass models for 175 disk galaxies with spitzer photometry and accurate rotation curves. *The Astronomical Journal*, 152:157, 2016. doi: 10.3847/0004-6256/152/6/157. URL https://arxiv.org/abs/1606.09251.

Federico Lelli, Stacy S. McGaugh, James M. Schombert, Harry Desmond, and Harley Katz. The baryonic tully–fisher relation for different velocity definitions and implications for galaxy angular momentum. *Monthly Notices of the Royal Astronomical Society*, 484:3267–3278, 2019. doi: 10.1093/mnras/stz205. URL https://arxiv.org/abs/1901.05966.

Tony Lindeberg. Feature detection with automatic scale selection. *International Journal of Computer Vision*, 30:79–116, 1998. doi: 10.1023/A:1008045108935. URL https://link.springer.com/article/10.1023/A:1008045108935.

Timothy Lingard, Karen L. Masters, Coleman Krawczyk, et al. Galaxy zoo builder: Morphological dependence of spiral galaxy pitch angle. *Monthly Notices of the Royal Astronomical Society*, 504:3364–3374, 2021. doi: 10.1093/mnras/stab1072. URL https://arxiv.org/abs/2105.04500.

Timothy K. Lingard, Karen L. Masters, Coleman Krawczyk, et al. Galaxy zoo builder: Four component photometric decomposition of spiral galaxies guided by citizen science. *The Astrophysical Journal*, 900:178, 2020. doi: 10.3847/1538-4357/ab9d83. URL https://arxiv.org/abs/2006.10450.

Karen L. Masters, Coleman Krawczyk, Shoaib Shamsi, et al. Galaxy zoo: 3D—crowdsourced bar, spiral, and foreground star masks for MaNGA target galaxies. *Monthly Notices of the Royal Astronomical Society*, 507:3923–3935, 2021. doi: 10.1093/mnras/stab2282. URL https://arxiv.org/abs/2108.02065.

Masanori Miyamoto and Ryuzaburo Nagai. Three-dimensional models for the distribution of mass in galaxies. *Publications of the Astronomical Society of Japan*, 27:533–543, 1975. doi: 10.1093/pasj/27.4.533. URL https://academic.oup.com/pasj/article/27/4/533/8122399.

Johanna Pasquet, Emmanuel Bertin, Marie Treyer, Stéphane Arnouts, and Dominique Fouchez. Photometric redshifts from SDSS images using a convolutional neural network. *Astronomy and Astrophysics*, 621:A26, 2019. doi: 10.1051/0004-6361/201833617. URL https://arxiv.org/abs/1806.06607.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, et al. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011. URL https://jmlr.org/papers/v12/pedregosa11a.html.

John C. Platt. Probabilities for SV machines. In Alexander J. Smola, Peter L. Bartlett, Bernhard Schölkopf, and Dale Schuurmans, editors, *Advances in Large-Margin Classifiers*, pages 61–74. MIT Press, 2000. doi: 10.7551/mitpress/1113.003.0008. URL https://doi.org/10.7551/mitpress/1113.003.0008.

David R. Roberts, Volker Bahn, Simone Ciuti, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40:913–929, 2017. doi: 10.1111/ecog.02881. URL https://doi.org/10.1111/ecog.02881.

James Schombert, Stacy McGaugh, and Federico Lelli. Using the baryonic tully–fisher relation to measure H_0. *The Astronomical Journal*, 160:71, 2020. doi: 10.3847/1538-3881/ab9d88. URL https://arxiv.org/abs/2006.08615.

Donald J. Schuirmann. A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15:657–680, 1987. doi: 10.1007/BF01068419. URL https://link.springer.com/article/10.1007/BF01068419.

John Y. H. Soo, Bruno Moraes, Benjamin Joachimi, et al. Morpho-z: improving photometric redshifts with galaxy morphology. *Monthly Notices of the Royal Astronomical Society*, 475:3613–3632, 2018. doi: 10.1093/mnras/stx3201. URL https://arxiv.org/abs/1707.03169.

Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel J. Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL https://arxiv.org/abs/1904.06019.

R. Brent Tully and J. Richard Fisher. A new method of determining distances to galaxies. *Astronomy and Astrophysics*, 54:661–673, 1977. URL https://articles.adsabs.harvard.edu/pdf/1977A%26A....54..661T.

Donald G. York, J. Adelman, John E. Anderson, et al. The sloan digital sky survey: Technical summary. *The Astronomical Journal*, 120:1579–1587, 2000. doi: 10.1086/301513. URL https://arxiv.org/abs/astro-ph/0006396.